Hidden Markov Models: Forward and Viterbi Algorithms

COM3031 Probabilistic Machine Learning

1 Forward Algorithm (Weather Example)

Hidden states:

$$S = \{\text{Rainy}, \text{Sunny}\}$$

Observation sequence:

$$O = (\text{Walk}, \text{Shop}, \text{Clean})$$

Initial distribution:

$$\pi = \begin{bmatrix} 0.6 & 0.4 \end{bmatrix}$$

Transition matrix:

$$A = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

Emission matrix:

$$B = \begin{array}{c|ccc} & Walk & Shop & Clean \\ \hline Rainy & 0.1 & 0.4 & 0.5 \\ Sunny & 0.6 & 0.3 & 0.1 \end{array}$$

Forward Variables

We define the forward variable:

$$\alpha_t(i) = P(o_1, \ldots, o_t, s_t = i \mid \lambda)$$

This represents the probability of observing the partial sequence $o_1, \ldots, o_t$ and ending in state i.

Forward Algorithm

Initialization

$$\alpha_1(j) = \pi_j b_j(o_1)$$

Recursion

$$\alpha_t(j) = \sum_i \alpha_{t-1}(i) a_{ij} b_j(o_t)$$

Termination

$$P(O|\lambda) = \sum_i \alpha_T(i)$$

Step 1: Initialization

For the first observation $o_1 = Walk$

$$\alpha_1(R) = \pi_R b_R(Walk) = 0.6 \times 0.1 = 0.06$$

$$\alpha_1(S) = \pi_S b_S(Walk) = 0.4 \times 0.6 = 0.24$$

Step 2: Recursion

Observation $o_2 = Shop$

$$\alpha_2(R) = [\alpha_1(R)a_{RR} + \alpha_1(S)a_{SR}]b_R(Shop) = [(0.06)(0.7) + (0.24)(0.4)] \times 0.4$$
$$= (0.042 + 0.096) \times 0.4 = 0.0552$$
$$\alpha_2(S) = [\alpha_1(R)a_{RS} + \alpha_1(S)a_{SS}]b_S(Shop) = [(0.06)(0.3) + (0.24)(0.6)] \times 0.3$$
$$= (0.018 + 0.144) \times 0.3 = 0.0486$$

Step 3: Recursion

Observation $o_3 = Clean$

$$\alpha_3(R) = [\alpha_2(R)a_{RR} + \alpha_2(S)a_{SR}]b_R(Clean) = [(0.0552)(0.7) + (0.0486)(0.4)] \times 0.5$$
$$= (0.03864 + 0.01944) \times 0.5 = 0.02904$$
$$\alpha_3(S) = [\alpha_2(R)a_{RS} + \alpha_2(S)a_{SS}]b_S(Clean) = [(0.0552)(0.3) + (0.0486)(0.6)] \times 0.1$$
$$= (0.01656 + 0.02916) \times 0.1 = 0.004572$$

Step 4: Termination

The likelihood of the observation sequence is:

$$P(O|\lambda) = \alpha_3(R) + \alpha_3(S) = 0.02904 + 0.004572 = 0.0336$$

2 Viterbi Algorithm (Weather Example)

Goal (Decoding)

Find the most likely hidden state sequence:

$$s^{\star}_{1:T} = \arg\max_{s_{1:T}} P(s_{1:T} \mid o_{1:T}, \lambda) = \arg\max_{s_{1:T}} P(s_{1:T}, o_{1:T} \mid \lambda)$$

Viterbi Variables

$$v_t(i) = \max_{s_{1:t-1}} P(s_{1:t-1}, s_t = i, o_{1:t} \mid \lambda)$$

$$\psi_t(i) = \arg\max_{j} [v_{t-1}(j)a_{ji}]$$

Initialization ($o_1 = Walk$)

$$v_1(R) = \pi_R b_R(W) = 0.6 \times 0.1 = 0.06$$
$$v_1(S) = \pi_S b_S(W) = 0.4 \times 0.6 = 0.24$$

Recursion

For $t = 2, \ldots, T$ and each state $i \in \{R, S\}$:

$$v_t(i) = b_i(o_t) \max_{j \in \{R,S\}} \big[v_{t-1}(j) a_{ji}\big], \qquad \psi_t(i) = \arg \max_{j \in \{R,S\}} \big[v_{t-1}(j) a_{ji}\big].$$

For $o_2 =$ Shop

$$v_2(R) = b_R(Shop) \max\big[v_1(R) a_{RR},\ v_1(S) a_{SR}\big] = 0.4 \max(0.06 \cdot 0.7,\ 0.24 \cdot 0.4) = 0.0384,$$

$$\psi_2(R) = \arg\max\big(0.06 \cdot 0.7,\ 0.24 \cdot 0.4\big) = S.$$

$$v_2(S) = b_S(Shop) \max\big[v_1(R) a_{RS},\ v_1(S) a_{SS}\big] = 0.3 \max(0.06 \cdot 0.3,\ 0.24 \cdot 0.6) = 0.0432,$$

$$\psi_2(S) = \arg\max\big(0.06 \cdot 0.3,\ 0.24 \cdot 0.6\big) = S.$$

For $o_3 =$ Clean

$$v_3(R) = b_R(Clean) \max\big[v_2(R) a_{RR},\ v_2(S) a_{SR}\big] = 0.5 \max(0.0384 \cdot 0.7,\ 0.0432 \cdot 0.4) = 0.01344,$$

$$\psi_3(R) = \arg\max\big(0.0384 \cdot 0.7,\ 0.0432 \cdot 0.4\big) = R.$$

$$v_3(S) = b_S(Clean) \max\big[v_2(R) a_{RS},\ v_2(S) a_{SS}\big] = 0.1 \max(0.0384 \cdot 0.3,\ 0.0432 \cdot 0.6) = 0.002592,$$

$$\psi_3(S) = \arg\max\big(0.0384 \cdot 0.3,\ 0.0432 \cdot 0.6\big) = S.$$

Termination

$$s_3^\star = \arg\max(v_3(R), v_3(S)) = R$$

Backtracking:

$$s_2^\star = R, \qquad s_1^\star = S$$

$$s_{1:3}^\star = (S, R, R)$$

$$\text{Sunny} \to \text{Rainy} \to \text{Rainy}$$